

Basic Concepts

How epidemiologists reason about cause and effect — and how they know when they might be wrong

—— Based on “Basic Concepts” by Kenneth J. Rothman & Sander Greenland

WHERE WE'RE HEADED

Five Big Questions This Chapter Answers



What does it mean to say something “causes” disease?

The causal-pie model



How do we ever know we've found a real cause?

Philosophy of scientific inference



How much disease is there, exactly?

Incidence, prevalence, person-time



How much does an exposure change that?

Effects, ratios, attributable fractions



What can quietly distort the answer?

Confounding and bias

CAUSATION · A STARTING DEFINITION

What Do We Mean by “a Cause”?

We define a cause of disease as something that came before the disease and without which it either would not have happened at all, or would not have happened when it did. That definition has a quiet but important consequence: a single cause is rarely, by itself, **enough** to produce disease. It is usually just one necessary piece of a larger set of conditions.

A familiar example: what makes a light turn on?



A working bulb



Intact wiring



Live current

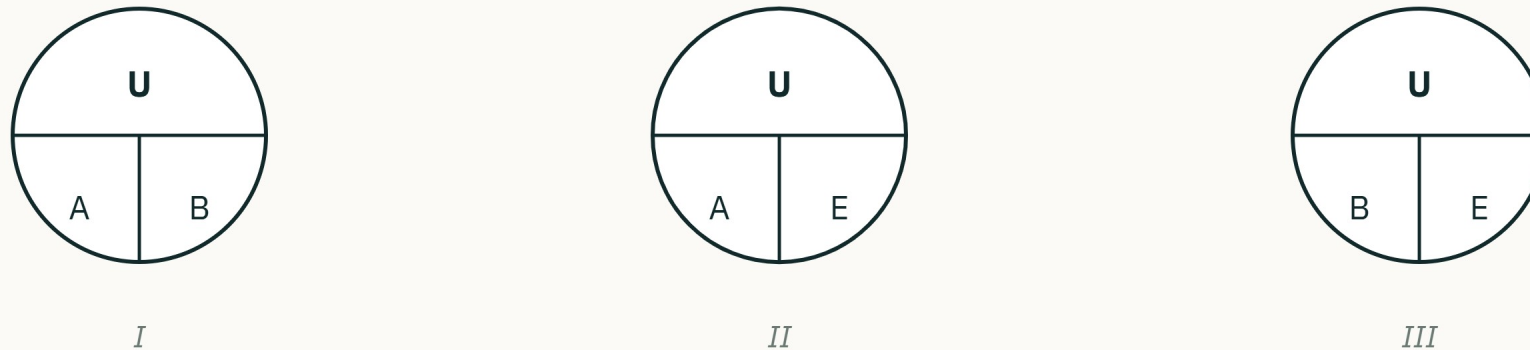


A flipped switch

Flipping the switch looks like “the” cause, but the bulb, the wiring, and the current are equally necessary — take any one away and the light never comes on.

The Causal Pie: How Causes Combine

Picture a complete cause of disease as a pie. Each slice is a **component cause** — one necessary piece. Only when every slice is in place is the pie whole, or “**sufficient**”: the disease is now guaranteed to begin.



Three different sufficient causes of the **same** disease. “U” happens to sit in all three — making it a **necessary cause**, since no case can occur without it. “A,” “B,” and “E” each appear in only two; any one component can take part in one mechanism, several, or all of them.

APPLYING THE MODEL

Why Smoking Alone Isn't "Enough"

Tobacco smoking is a cause of lung cancer, but on its own it is not a **sufficient** cause: most heavy smokers never develop lung cancer. "Smoking" also has to be pinned down — type of smoke, filtered or not, how deeply it's inhaled, and for how long — before it even counts as one well-defined slice of the pie.

The missing slices are unknown, not absent

Some heavy smokers chain-smoke for decades without developing cancer; others seem "primed" by circumstances we can't yet measure. Everyone in the category gets the same average risk only because we can't see who already has the other components in place.

Knowledge advances by finding more slices

Heavy smokers as a group are often said to face roughly a 10% lifetime risk of lung cancer. But smokers who are **also** exposed to asbestos face a higher risk than those who aren't — asbestos exposure was hiding inside that average all along.

TWO MORE PROPERTIES OF THE MODEL

When Causes Depend on Their Partners

A cause looks “strong” or “weak” depending on its partners

The size of a factor’s measured effect depends on how common its remaining causal partners — together called its **causal complement** — are in the population, not on anything different about its biology. The same component with a common causal complement will appear to drive a lot of disease; with a rare one, it will look almost harmless.

Two causes can interact even decades apart

A head injury can permanently disturb someone’s balance. Years later, a walk along an icy path causes a fall and a broken hip. Both events are components of the same causal mechanism — they interacted biologically, even though one acted years before the other.

Head injury disturbs balance



Years pass, unnoticed



Icy path, a fall, a broken hip

A COUNTER-INTUITIVE CONSEQUENCE

Why the Causes of a Disease Can Add Up to More Than 100%

Since a sufficient cause is a whole constellation of components, a single case of disease almost always has more than one thing that can honestly be called “a cause.” That breaks a very natural assumption: that the shares of disease attributed to all its different causes should add up to 100%.

A famous “anomaly” that wasn’t one

In their widely cited work *The Causes of Cancer*, Doll and Peto published a table of estimated cancer causes whose shares summed to nearly 100%. Because any single case could have more than one cause, they called this a “difficulty,” or an “anomaly.”

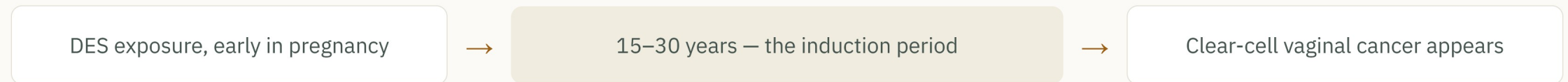
It isn’t an anomaly at all

No event has just one cause. The total share of disease attributable to **all** of its causes, added together, has no ceiling of 100% — it can run to infinity. Only the share attributable to one single, specific cause can never exceed 100%.

REAL EXAMPLE · DES AND VAGINAL CANCER

The Delay Between Cause and Disease

The **induction period** is the time between a cause acting and the disease actually beginning. A classic, sobering example: female fetuses exposed to the drug DES early in pregnancy went on to develop a rare vaginal cancer — but usually not until ages 15 to 30.



A separate clock: the latent period

Induction is the wait until disease **begins**. The **latent period** is the different wait until it is **detected**. Better screening can shrink the latent period, but it can never shrink the induction period — that clock stops the moment the disease actually starts.

Induction time belongs to a pair, not a disease

DES has this 15–30 year induction time for vaginal cancer specifically; another component cause of that same cancer, acting later in the mechanism, could have a far shorter induction time. And direction matters: hastening an event makes something a cause; postponing it makes something preventive.

Source: Rothman (1981); the diethylstilbestrol (DES) – clear cell adenocarcinoma relationship.

Inductivism: Building Laws From Observations

Modern science took shape in the 1600s with the idea, championed by Francis Bacon, that a scientist could make careful observations and then **induce** — generalize upward to — a natural law. It was a real advance over relying on faith, ancient texts, or authority.

The textbook example: Jenner and smallpox

Edward Jenner noticed that milkmaids, commonly exposed to cowpox, rarely caught smallpox. From that repeated observation, he induced a general law: cowpox infection confers immunity to smallpox — the foundation of vaccination.

Inductivist thinking reached a peak in John Stuart Mill's 1843 canons of inference, which evolved into criteria for weighing evidence that epidemiologists still lean on today.

THE CRACK IN INDUCTIVISM

The Trap of Mistaking Correlation for Confirmation

The philosopher David Hume pointed out a fatal flaw: an inductive argument has no real logical force. Saying “no smallpox has followed cowpox so far, so it never will” is just an assumption that the future repeats the past — not a proof of anything.

A common trap: “confirming” a hypothesis with a matching observation

Consider: *if* sewer service causes heart disease, *then* heart disease rates should be highest where sewer service is available. Rates **are** highest there. Therefore, sewer service causes heart disease.

The logic is unsound — there are many other ways both things could be true at once. Here, economic development could raise **both** sewer access and heart disease rates, with no effect of one on the other at all. The association would then be **spurious**, or confounded.

The same trap has a name when it’s about timing alone: the *post hoc* fallacy — assuming the rooster’s crow makes the sun rise, simply because crowing always comes first.

A RIVAL PHILOSOPHY

Refutationism: Try to Disprove, Rather Than Prove

Karl Popper's answer to Hume: consistency with observations can never **prove** a hypothesis — there is always some alternative hypothesis that fits the same data. A single clean observation that contradicts a hypothesis can seem to **refute** it outright. But observations themselves depend on assumptions and can carry unnoticed bias, so even an apparent refutation is only as solid as the observation behind it — logical certainty about a theory, true or false, is never fully within reach.



Stop the rooster

If the sun still rises after you silence the rooster, you've refuted the idea that crowing is necessary for sunrise — with a single trial.



Boiling water

No number of repeats in an open flask can prove water always boils at 100°C. But one attempt in a closed flask, or at altitude, refutes it immediately.

By this view, science advances through “conjecture and refutation”: propose a hypothesis, then go looking for the observation that could break it — rather than collecting evidence that merely agrees with a favorite theory.

A THIRD PHILOSOPHY

Bayesianism: Updating Belief Instead of Chasing Certainty

Classical logic can only draw a valid conclusion from premises you are 100% certain of — and in health research, we are never fully certain of anything. Bayesian reasoning offers a way around this: start from a stated degree of belief (a **prior**), then update it in light of the data to reach a **posterior** degree of belief.

Why the starting point matters: television and violence

Consider the claim: *if* television causes homicide, *then* homicide rates should be highest where televisions are most common. That premise is itself doubtful — homicide rates can be low where televisions are common, for entirely unrelated economic reasons. Two researchers who start with different beliefs about that premise can validly reach different conclusions from the very same data.

This dependence on a starting point is why Bayesian conclusions are called **subjective** — not because truth itself is subjective, but because the certainty you end with depends on the certainty you honestly started with.

A SHARED LIMITATION OF ALL SCIENCE

Proof Is Impossible — in Any Science

Epidemiologists are often told that, unlike “real” experimental sciences, they can never prove causation. But proof is impossible in empirical science generally — plate tectonics, the evolution of species, planets orbiting other stars, and the health effects of smoking were all established without ever being definitively “proven.”

Even experiments don’t deliver certainty

Randomized trials and laboratory experiments can still be controversial, mutually inconsistent, or impossible to reproduce. The 1989 “cold fusion” episode — a widely announced experimental discovery that could not be replicated — shows that neither physical nor experimental science is immune to this problem.

“I cannot give any scientist of any age better advice than this: the intensity of the conviction that a hypothesis is true has no bearing on whether it is true or not.”

— Peter Medawar, 1979. The reverse holds too: conviction that a hypothesis is **false** has no bearing on whether it is false.

REAL EXAMPLE · TOXIC SHOCK SYNDROME

Toxic Shock: Two Theories, One Decisive Clue

When toxic shock syndrome first appeared among tampon users, two rival theories explained it — and, crucially, they made **opposite** predictions about one specific behavior: how often a woman changed her tampon.

Theory 1: chemical toxin

The tampon itself exposes the body to a toxic chemical. Predicts: **more frequent changing** → **more exposure** → **higher risk**.

Theory 2: culture medium

The tampon lets bacteria grow, which then produce the toxin. Predicts: **more frequent changing** → **less time to grow** → **lower risk**.

What the data showed

Infrequent tampon changing was associated with a **greater** risk of toxic shock — the pattern the culture-medium theory predicted. The chemical-toxin theory was refuted.

Source: an example of testing two competing causal hypotheses against each other via one specific, opposite prediction.

REAL EXAMPLE · ESTROGEN AND ENDOMETRIAL CANCER

Estrogen and Endometrial Cancer: Ruling Out a Rival Explanation

Women taking replacement estrogen showed a much higher rate of endometrial cancer. A rival, noncausal theory proposed that estrogen causes bleeding, which sends women to the doctor sooner — so their cancer is simply **found** more often, not caused more often.

If it's a real, causal effect

Risk should keep climbing with higher dose, frequency, and duration of use — the pattern typical of a true carcinogen.

If it's only detection bias

Risk should look highest among **short-term** users, since symptom-driven checkups happen soon after estrogen use begins.

What the data showed

The increased risk was just as strong in long-term users as in short-term users. Detection bias was refuted as the explanation for all but a small fraction of cases — it would also have required an implausibly large hidden reservoir of undetected cancer in the general population.

A WIDELY USED CHECKLIST • USED WITH CAUTION

Hill's Nine Considerations

In 1965, Austin Bradford Hill proposed nine aspects of an association worth weighing when judging whether it might be causal. He was careful never to call them “criteria.”

1 • Strength

A strong association is harder to explain away

2 • Consistency

Seen repeatedly, by different people, places, times

3 • Specificity

One exposure, one specific outcome

4 • Temporality

The cause must come before the effect — the one truly required

5 • Biological gradient

More exposure tracks with more disease

6 • Plausibility

A believable mechanism exists, when we can check

7 • Coherence

Fits what's otherwise known about the disease

8 • Experimental evidence

Supportive trial results, where any exist

9 • Analogy

Similar exposures are known to cause similar effects

Hill himself stressed that, apart from temporality, none of these is strictly necessary or sufficient. Used as a rigid checklist to defend a favorite theory, they can cloud judgment rather than sharpen it.

MEASURES OF DISEASE FREQUENCY

Four Ways to Measure How Much Disease There Is

Simply counting cases isn't enough — we also need to know over what time, and out of how many people. Epidemiology leans on four basic measures to make that precise.



Incidence time

The moment new disease occurs in one person



Incidence rate

New cases per unit of time spent at risk



Incidence proportion

The share of people who develop it in a given period



Prevalence

The share of people who currently have it, right now

Incidence Time: Measured From What?

Before we can time an event, we need a starting point — a “zero time.” It might be birth, so we measure in age; it might be the start of a treatment or exposure; or it might simply be a calendar date. **Incidence time** is the span from that zero time to when the event happens.

Same event, two clocks

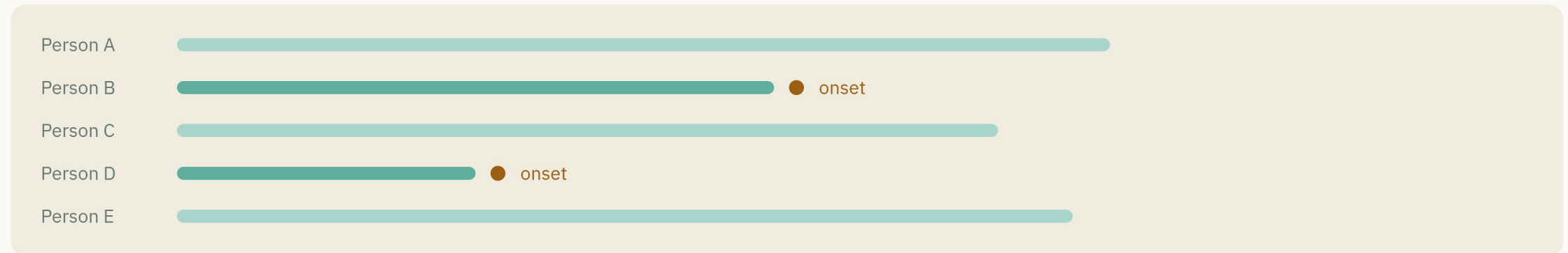
A man has his first heart attack in 1990 at age 50. In calendar time, his incidence time is 1990. In age time, it's 50. Neither is more “correct” — they're just different reference points.

What if it never happens?

A woman who has a hysterectomy in 1990 without ever having had endometrial cancer simply has an incidence time for that cancer known only to be later than 1990.

Incidence Rate: New Cases per “Person-Time”

Not every event of interest is inevitable, and people are observed for different lengths of time. The **incidence rate** divides new cases by the total **person-time** — the sum of however long each person was followed and at risk.



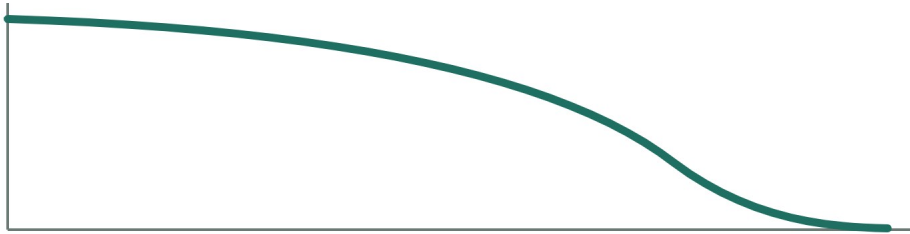
Two of five people developed the disease. The incidence rate divides that count by the total time everyone spent under observation — not by the number of people — so it carries units like “cases per person-year.” A case only belongs in that count if the person was still contributing eligible time at the moment it happened — for a first disease occurrence, that person then stops contributing time; a later recurrence needs its own, separate risk set. (Timeline above is an author-created illustration.)

TWO KINDS OF POPULATIONS TO STUDY

Closed and Open Populations

Closed population

No one joins after the start; people only leave through death. Follow 1,000 people long enough and the group shrinks steadily toward zero.



Population size (up) over time (across) — author-created illustration based on Figure 3

Open population

People can also join — by birth, immigration, or aging into an age group — while others still leave. If entries and exits balance, the population is in a **steady state**.



People continually enter and leave; the total stays flat — author-created illustration based on Figure 4

A RATE IS NOT A PROPORTION

Reading an Incidence Rate Correctly

It has no ceiling of 1.0

Because it's measured per unit of time, not as a share of people, a rate can exceed 1.0. If 100 people died after an average of just 3.65 days each, the death rate would be 100 per person-year — far above “1.”

The number alone means nothing

100 cases per person-year is exactly the same rate as 8.33 per person-month, 1.92 per person-week, or 0.27 per person-day. Always state the time unit.

The “waiting time” shortcut — and where it breaks

In a population with no migration, the reciprocal of the death rate approximates the average life span. The 1977 U.S. death rate of 0.0088 per year implies about 114 years — but actual life expectancy that year was 73. The gap reflects immigration, plus a failure of the steady-state assumption itself — the U.S. was not truly a closed, steady-state population.

Incidence Proportion: Risk, Made Population-Wide

Over a fixed time interval in a closed population, the **incidence proportion** — often just called **risk** — is simply the share of people who become a case.

10%

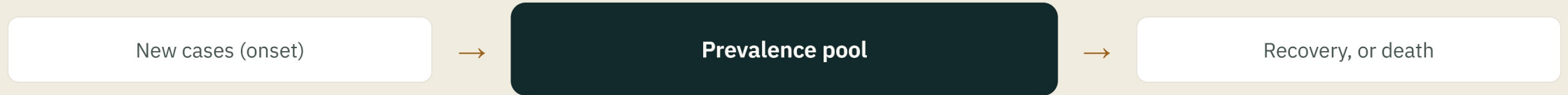
10 of 100 people, over 5 years

A proportion is meaningless without its time period: 10% over 5 years describes a very different disease than 10% over 5 days.

Its natural complement is the **survival proportion** — the share who stay disease-free. The **incidence odds**, the ratio of the two, comes out close to the risk itself whenever the risk is small. A familiar example is case-fatality “rate” — despite its name, it’s really an incidence proportion, and like any proportion it only means something once its time horizon is stated.

Prevalence: A Snapshot, Not a Flow

Unlike the other three measures, prevalence isn't about new events at all — it's the share of a population that **currently has** the disease, at one point in time. The people with disease at that moment form the **prevalence pool**.



People leave the pool by recovering, dying, or emigrating — which is why a disease with a large incidence rate can still have a low prevalence, if it kills or resolves quickly.

WHEN TO REACH FOR PREVALENCE

What Prevalence Can — and Can't — Tell Us

Think of prevalence like the water level in a sink: it rises with how fast the tap runs (incidence) and how long the water lingers before it drains (duration). A treatment that only keeps people sick **longer** — without changing how often the disease starts — will raise prevalence, even though it did nothing to cause more disease. In a stable population with no migration this is precise: prevalence odds equal the incidence rate times the average disease duration, and when prevalence is low, prevalence itself is approximately incidence times duration.

That's why prevalence is rarely the right tool for finding causes — but it is often the only practical option:

Birth defects

Measured at birth — itself a prevalence — since many affected pregnancies end in miscarriage before the defect could ever be counted as an incident case.

Silent infections

HIV seroprevalence surveys are used precisely because the moment of infection so often passes unnoticed.

Planning health resources

For clinics and budgets, what matters is how many people need care right now — prevalence, not incidence, drives that burden.

MEASURES OF EFFECT

What Epidemiologists Mean by “Effect”

Sense 1: what kind

HIV infection is “an effect” of needle sharing — simply naming which outcome a cause can lead to.

Sense 2: how much

The actual amount of change a factor causes in a population’s disease frequency — a number, not just a label.

0.1% → 1%

For drug users, the effect of needle sharing (versus not) is to raise the one-year risk of HIV infection roughly tenfold — from about 0.1% to about 1%.

A THOUGHT EXPERIMENT

Measuring an Effect: A Thought Experiment

Picture a group of smokers followed for several years. Now imagine two different histories for that **same** group: mail every member a smoking-cessation brochure, or mail no one at all. Comparing the two histories reveals the effect of the mailing.

If everyone got the mailing

A certain lung-cancer incidence rate would result.

If no one got it

A different incidence rate would result instead.

The **gap** between those two rates is the **causal rate difference**. The same comparison done with incidence proportions instead gives the **causal risk difference**. Both are “absolute” measures — they compare by subtracting.

ONE COMPARISON, SEVERAL SCALES

The Same Causal Comparison, Read Different Ways

Every measure below compares the very same two hypothetical histories from the thought experiment — it’s only the arithmetic, and the disease-frequency measure it’s built from, that changes.

Frequency measure	Subtracting (absolute)	Dividing (relative)
Incidence rate	Causal rate difference	Causal rate ratio
Incidence proportion (risk)	Causal risk difference	Causal risk ratio
Disease-free time	Difference in disease-free time	Ratio of disease-free time
Incidence odds	—	Causal odds ratio

The chapter develops all of these as named measures — not just the rate and risk differences most often taught first.

THE IDEA UNDERNEATH EVERY EFFECT MEASURE

Comparing to What Didn't Happen

“The effect of X” is meaningless on its own — it only makes sense compared **against** some other condition. One mailing versus no mailing is one effect; one mailing versus four mailings is an entirely different comparison.

Why it's called “counterfactual”

A population can only actually experience one condition at a time. So at least one side of any effect comparison is, by definition, contrary to fact — a condition that could have happened, but didn't.

Which means effects must be estimated, not observed

Since we can never directly observe the missing condition, we predict it using outside information — data from a different but comparable group. That outside data is only ever an ingredient in the estimate, never the effect itself.

A SUBTLETY WORTH KNOWING

When the Effect Isn't the Same for Everyone

Split a population into groups — say, by sex — and an effect measure can look different in each group. That’s called **effect-measure modification**. A worked, illustrative example shows something surprising:

Group (illustrative)	Exposed	Unexposed	Difference	Ratio
Males	0.50	0.20	0.30	2.5
Females	0.10	0.04	0.06	2.5
Males, if their risk were 0.26 instead	0.26	0.20	0.06	1.3

With the top two rows, the ratio is uniform across sexes (2.5) but the difference isn’t. Change just one number, and the difference becomes uniform (0.06) instead — but now the ratio isn’t. When a real effect is present, at most one of the two ways of measuring it can stay uniform across groups.

Source: a worked illustrative example of effect-measure modification.

TWO LENSES ON THE SAME CHANGE

Relative and Absolute Effects Tell Different Stories

An investment that grows by \$1,000 in a year sounds useful — but how good was it, really? That depends entirely on where it started.

\$5,000 → \$6,000

\$1,000

absolute gain

the same \$1,000, relative to its start

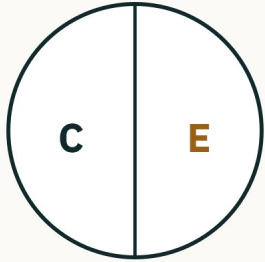
20%

relative gain

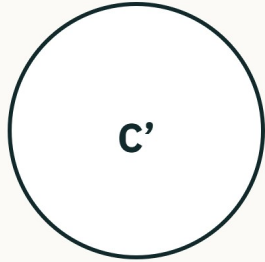
Disease effects work the same way. The very same absolute effect can look huge or trivial in relative terms — and the same relative effect can hide very different absolute effects — depending on how common the outcome was to begin with.

A SHARPER QUESTION THAN “HOW BIG IS THE EFFECT”

Not Every Exposed Case Was Caused by the Exposure



Mechanisms that include
the exposure



Mechanisms that don't need it at all

A smoker who develops lung cancer might have gotten it through a mechanism that never involved smoking at all — say, one built entirely from asbestos exposure. For that case, the smoking was incidental. Ordinary incidence data alone can't tell us which exposed cases belong to which circle. (Diagram is an author-created illustration.)

A COSTLY MIX-UP

The Attributable-Fraction Family — and a Costly Mix-Up

Researchers have proposed several “attributable fraction” measures — the rate fraction and the excess fraction among them. They sound interchangeable, but they are mathematically different quantities, and neither generally equals the true **etiologic fraction**: the share of exposed cases where the exposure actually played a causal role.

Why this matters outside the classroom

Courts and regulators often want the “probability of causation” for one particular person’s case. But the chapter defines it as the probability that exposure caused disease in a randomly selected exposed case — a population idea, and treating it as a fact about one individual is exactly the translation the authors warn against. That estimate can be considerably smaller than the true etiologic fraction, even for a rare disease.

A related, population-level idea

The **population attributable fraction** asks how much disease would disappear if the exposure were removed from the whole population. The **impact fraction** asks the same question for a partial reduction, not a complete one. A **preventable fraction** mirrors this for an exposure that lowers risk instead of raising it.

A DISTINCTION WORTH KEEPING STRAIGHT

An Association Isn't Automatically an Effect

An **effect** compares the same population under two conditions, one of them counterfactual. A **measure of association** compares two occurrence numbers from two **different** populations, or the same population at two different times — and that difference is not automatically an effect.

Different people

Comparing cancer rates between men and women in Canada is a measure of association with sex — not the “effect” of being male, since it’s far from clear what it would even mean for a specific woman to have instead “been a man.”

Different times

Comparing a town’s cavity rate before and after water fluoridation compares two different time periods, with possibly-different people and circumstances — not necessarily the true effect of the fluoridation itself.

That fluoridation comparison is the natural way to **estimate** an effect when the real counterfactual can’t be observed — but whenever the substitute isn’t a fair stand-in, the estimate goes wrong. That’s exactly what confounding is, covered next.

REAL EXAMPLE · WATER FLUORIDATION AND CAVITIES

Confounding: When the Stand-In Isn't Fair

To measure fluoridation's true effect, we'd compare today's cavity rate to the rate that **would have occurred today** had fluoridation never been introduced. We can't observe that missing number, so we substitute the rate from before fluoridation began instead.

What we actually want

Today, with fluoridation, versus today, without it (counterfactual, unobservable)

≠

What we can actually get

Today, with fluoridation, versus **before** fluoridation, a different time period entirely

Whenever that substitute rate doesn't equal the true, unobservable counterfactual, the comparison is said to be **confounded** — and it will overstate or understate fluoridation's real effect.

CONTINUING THE FLUORIDATION EXAMPLE

Confounders, and How Stratification Fixes Them

A **confounder** is a factor that would have changed the outcome on its own — even with no change in the exposure being studied — and so distorts the comparison.

Suppose dental-hygiene programs also launched that year

If schools rolled out hygiene education around the same time as fluoridation, cavities would have declined somewhat **anyway**. The before/after comparison now overstates fluoridation's real effect — the hygiene programs are confounding it.

The fix: stratification

Compare cavity rates only within schools that share the same hygiene-program status — for instance, only among schools that never introduced one. Within that slice, the confounder can no longer explain away the comparison.

REAL EXAMPLE · THE SMOKY ATOMIC TEST COHORT

Self-Selection Bias: Who Chooses to Be Counted

Selection bias, broadly, happens whenever the process of participation or selection itself distorts the exposure–disease relationship — because the people actually studied differ, in some relevant way, from the full population that was theoretically eligible. Self-selection and diagnostic bias, below, are two common ways that happens.

Studying leukemia among troops present at a 1957 nuclear test, investigators eventually established outcomes for 76% of the cohort — about 62% traced by the investigators themselves, and a further 14% who contacted researchers **on their own initiative** after publicity about the study.

76%

of the cohort had known outcomes

4 + 4

leukemia cases: self-referred and traced, equally

9–11

estimated true total, depending on the untraced 24%

Four cases among the small self-referred group, versus four among the much larger traced group, hints that self-referral was linked to the outcome — self-selection bias was **small but real** here.

Self-selection can also happen before anyone is even identified for study: working people, as a group, are routinely measured as healthier than the population as a whole — the “healthy-worker effect” — simply because unwell people are less likely to remain in or enter the workforce.

Source: Caldwell et al. (1980), the Smoky Atomic Test leukemia study.

Diagnostic Bias: When Suspicion Drives the Search

Early studies of oral contraceptives and venous thromboembolism relied on hospitalized patients. But by then, doctors already suspected a link — and that suspicion itself could distort the results.

The circular problem

A physician aware that a patient uses oral contraceptives, and aware of the suspected link, may investigate and diagnose a clotting event more thoroughly in that patient than in one who doesn't. That extra scrutiny — not any extra biology — can manufacture part of the very association being studied.

This is a form of selection bias that occurs before subjects are even identified for study — which patients get flagged, worked up, and diagnosed in the first place is already shaped by the hypothesis under test.

Differential Misclassification: Errors That Favor One Side

Once subjects are enrolled, errors can still creep into how exposure or disease gets measured. When the **size** of that error itself depends on a person's exposure or disease status, it's called **differential** misclassification.

Emphysema and smoking

Smokers seek medical attention more often, so emphysema — a disease that can go undiagnosed without a check-up — gets **found** more often among smokers. The apparent excess is partly just better detection, not necessarily more disease.

Recall bias in birth defects

A mother who just had a malformed baby searches her memory for possible causes far more carefully than a mother of a healthy baby. Her more thorough recall of infections, injuries, or drugs can manufacture an apparent effect on its own.

Nondifferential Misclassification: Even-Handed Errors Still Bias

When measurement error doesn't depend on the other variable, it's **nondifferential**. For a simple yes/no exposure, this usually pushes results **toward** the null — but “usually” is doing real work: it is not automatically safe to ignore.

Aspirin and heart attacks

Aspirin's protective effect only applies in a brief window before an event. Studies that instead ask about “ever use” blur that narrow window into a much broader, less precise measure — diluting a real effect toward zero.

Bendectin and birth defects

One study grouped many unrelated birth defects into a single “malformation” outcome. Lumping categories together like this can mask a real, narrower effect — such as one specific to heart defects — inside a diluted, unremarkable-looking average.

A similar logic applied to a spermicides-and-birth-defects study: because the imprecise exposure data was recorded before anyone knew the outcome, any resulting bias would most likely have hidden a real effect, not invented one.

REAL EXAMPLE · COFFEE, SMOKING, AND BLADDER CANCER

Misclassifying a Confounder Can Bias Either Way

Smoking is strongly linked to both coffee drinking and bladder cancer, making it a powerful confounder of any studied coffee–bladder cancer relationship. Controlling for it well matters enormously — and controlling for it well requires measuring it well.

The asymmetry that makes this dangerous

Misclassifying the **exposure** under study usually biases results toward the null — a predictable, “safer” direction. Misclassifying a **confounder** is different: because it weakens how well you can control for it, the resulting bias can run in **either** direction, and it is most dangerous exactly when the real coffee–cancer relationship is weak or nonexistent.

Even a crude smoking measure — a simple “ever smoked” versus “never smoked” — loses the detail needed to fully control for it, so some residual confounding by smoking is nearly unavoidable, however carefully it’s measured.

Three Habits That Distort Statistical Results

Beyond confounding and bias, a study's results can be misread even when everything else was done well. Three habits of mind, working together, are especially common:

Dichotomania

Squeezing a continuous, graded result down into a bare “significant” or “not significant.”

Nullism

Treating a study that failed to rule out “no effect” as if it had proven there is no effect.

Statistical reification

Forgetting that every statistical result rests on assumptions, and reading it as a direct fact about reality regardless.

In practice: a study with $p = 0.08$, or a confidence interval that just crosses the null, often gets reported as showing “no effect” — when the honest reading is simply that the data didn't rule the null out. Some methodologists now suggest reframing p-values as **compatibility measures** and confidence intervals as **compatibility intervals**, to keep this distinction from getting lost.

Together, these habits produce a systematic, dramatic understatement of uncertainty — and an overconfident reading of what a study actually showed.

WRAPPING UP · 2 OF 2

Epidemiologic inference is never certain — and it is more than statistics.

Every conclusion in this chapter — a cause identified, a rate calculated, an effect estimated — is a tentative best answer, always open to revision. What separates careful epidemiology from a bare statistical result is exactly the material covered here: a working model of causation, an honest philosophy of evidence, precise measures of disease and effect, and constant vigilance against confounding and bias.

SOURCE & ATTRIBUTION

Rothman, K.J. & Greenland, S. (2023). “Basic Concepts.”

In: *Handbook of Epidemiology*, W. Ahrens & I. Pigeot (eds), Springer Nature.

Prepared by Omokhodion Alfred Eriakha · September 18, 2026

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